



Double-null and limiters

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Outline

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General philosophy

The standard code expects boundaries only at the edges of the computational domain. When this is no longer true we have a 'complex' geometry. In that case, the code must be told where the boundaries are and what to apply where. The neighbourhood array computed by **b2ag** already contains information about the boundaries, in the following way:

East `leftix(ix,iy) = -2`

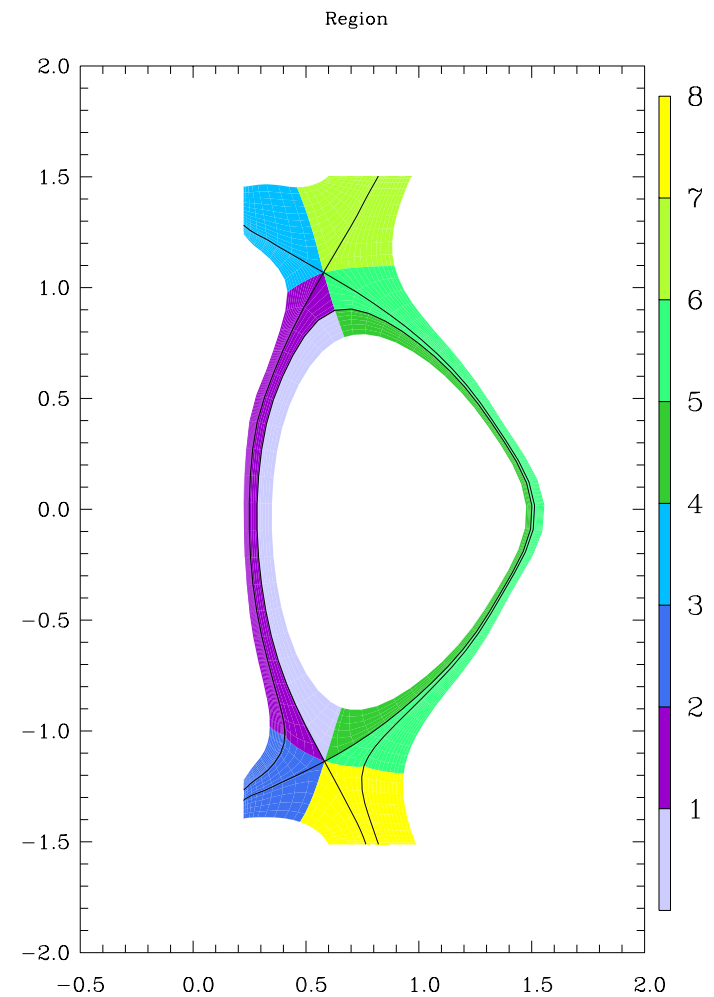
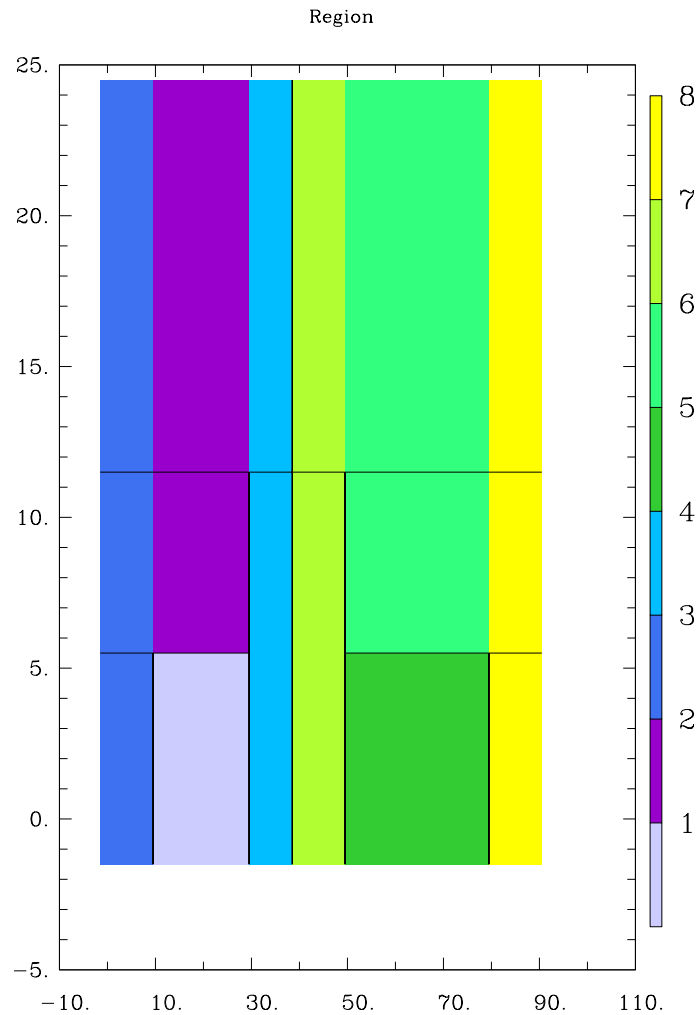
West `rightix(ix,iy) = nx+1`

North `topiy(ix,iy) = ny+1`

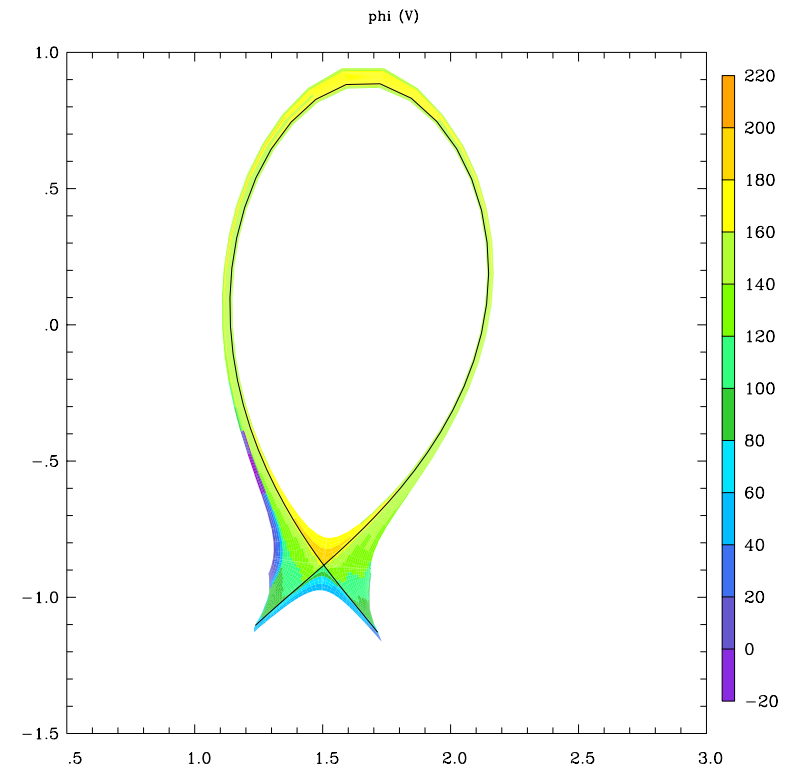
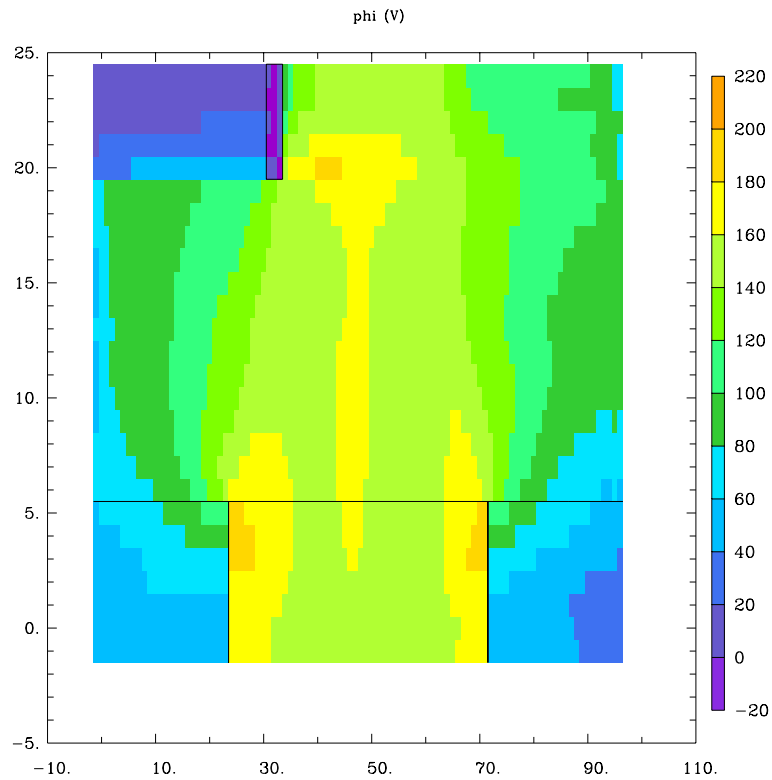
South `bottomiy(ix,iy) = -2`

The information about the details of the boundary conditions must be provided via the **b2.boundary.parameters** file. 'Unused' regions such as limiters will be defined as dead regions (**region(ix,iy,0) = 0**) and also declared in **b2.boundary.parameters**. **b2ag** makes sure that dead region cells are only connected within themselves and not mistakeable for boundary cells.

How B2 sees a double-null geometry



How B2 sees a limiter geometry





Necessary changes in the input files

In `b2mn.dat` and `b2ah.dat`, one must still correctly declare the boundaries at the edges of the domain. This means that the `cbreg` definitions now look like this:

```
*b2cmpb          boundary conditions
*cbregs
'south'          6 0.11111111 0.33333333 0.43333333 0.55555556 0.88888889 2.00
'north'          2 0.43333333 2.0
'west'           1          2.0
'east'           1          2.0
```

Note that there are 6 South boundaries (4 PFR halves and 2 core), 2 North boundaries (both SOLs), but only 1 East plate and 1 West plate as we can only here define the boundaries at the edges of the computational domain.

In a double-null, there are now 8 regions, and thus arrays in `b2.numerics.parameters` must be accordingly sized.

Double-null b2.boundary.parameters (part I)

&BOUNDARY

```

NBC=12,
BCCHAR=      'S',      'S',      'S',      'S',      'S',      'S',      'N',      'N',      'W',      'E',      'W',      'E',
BCPOS=       -1,       -1,       -1,       -1,       -1,       -1,      24,      24,      -1,      38,      39,      90,
BCSTART=      10,        0,      30,      50,      40,      80,       0,      40,      -1,      -1,      -1,      -1,
BCEND=        29,        9,      37,      79,      49,      89,      37,      89,      24,      24,      24,      24,
BCENE=         8,        4,        4,        8,        4,        4,        4,        4,        3,        3,        3,        3,
BCENI=         8,        4,        4,        8,        4,        4,        4,        4,        3,        3,        3,        3,
BCPOT=         0,        0,        0,        0,        0,        0,        0,        0,        3,        3,        3,        3,
ENEPAR=  0.11e6,    1.00,    1.00, 0.39e6,    1.00,    1.00,    1.00,    1.00,    0.90,    0.90,    0.90,    0.90,
ENIPAR=  0.11e6,    1.00,    1.00, 0.39e6,    1.00,    1.00,    1.00,    1.00,    2.50,    2.50,    2.50,    2.50,
POTPAR=   0.00,    0.00,    0.00,    0.00,    0.00,    0.00,    0.00,    0.00,    0.00,    0.00,    0.00,    0.00,
ENEPAR(2,2)=2*1e2,    ENEPAR(5,2)=4*1e2, ENIPAR(2,2)=2*1e2,    ENIPAR(5,2)=4*1e2, POTPAR(1,2)=12*0,
BCCON(0,1)   =    8,        1,  CONPAR(0,1,1) = 0e19,    2e19,
BCCON(0,2)   =   10,       10,  CONPAR(0,2,1) =-1.0e-2, -1.0e-3,
BCCON(0,3)   =   10,       10,  CONPAR(0,3,1) =-1.0e-2, -1.0e-3,
BCCON(0,4)   =    8,        1,  CONPAR(0,4,1) = 0e19,    2e19,
BCCON(0,5)   =   10,       10,  CONPAR(0,5,1) =-1.0e-2, -1.0e-3,
BCCON(0,6)   =   10,       10,  CONPAR(0,6,1) =-1.0e-2, -1.0e-3,
BCCON(0,7)   =   10,       10,  CONPAR(0,7,1) =-1.0e-2, -1.0e-3,
BCCON(0,8)   =   10,       10,  CONPAR(0,8,1) =-1.0e-2, -1.0e-3,
BCCON(0,9)   =    5,        3,  CONPAR(0,9,1) = 0.00,    0.00,
BCCON(0,10)  =    5,        3,  CONPAR(0,10,1)= 0.00,    0.00,
BCCON(0,11)  =    5,        3,  CONPAR(0,11,1)= 0.00,    0.00,
BCCON(0,12)  =    5,        3,  CONPAR(0,12,1)= 0.00,    0.00,

```



Double-null b2.boundary.parameters (part II)

```
BCMOM(0,1) = 4, 4, MOMP(0,1,1) = 0.0, 0.0, mompar(0,1,2) =1e5,1e5,
BCMOM(0,2) = 4, 4, MOMP(0,2,1) = 0.0, 0.0, mompar(0,2,2) =1e5,1e5,
BCMOM(0,3) = 4, 4, MOMP(0,3,1) = 0.0, 0.0, mompar(0,3,2) =1e5,1e5,
BCMOM(0,4) = 4, 4, MOMP(0,4,1) = 0.0, 0.0, mompar(0,4,2) =1e5,1e5,
BCMOM(0,5) = 4, 4, MOMP(0,5,1) = 0.0, 0.0, mompar(0,5,2) =1e5,1e5,
BCMOM(0,6) = 4, 4, MOMP(0,6,1) = 0.0, 0.0, mompar(0,6,2) =1e5,1e5,
BCMOM(0,7) = 4, 4, MOMP(0,7,1) = 0.0, 0.0, mompar(0,7,2) =1e5,1e5,
BCMOM(0,8) = 4, 4, MOMP(0,8,1) = 0.0, 0.0, mompar(0,8,2) =1e5,1e5,
BCMOM(0,9) = 1, 3, MOMP(0,9,1) = 0.0, 1.0, mompar(0,9,2) =0,0,
BCMOM(0,10) = 1, 3, MOMP(0,10,1)= 0.0, 1.0, mompar(0,10,2)=0,0,
BCMOM(0,11) = 1, 3, MOMP(0,11,1)= 0.0, 1.0, mompar(0,11,2)=0,0,
BCMOM(0,12) = 1, 3, MOMP(0,12,1)= 0.0, 1.0, mompar(0,12,2)=0,0,
GAMMAI=1.666666667, GAMMAE=0.5, LBNDUSR=F/
```



What does it all mean?

A boundary condition is given a direction (BCCHAR), position (BCPOS), beginning (BCSTART) and end (BCEND). Different types of b.c. are available, and they are given by the BCENE, BCENI, BCPOT, BCMOM and BCCON flags. The ENEPAR, ENIPAR, POTPAR, CONPAR, and MOMPARG arrays contain the values being imposed. The significance of the flags is explained in `b2stbc_phys`. The most important ones are (with some exceptions!):

- 1 Given constant
- 2 Given gradient
- 3 Sheath
- 4 Weak constant
- 5 Constant flux per unit area
- 6 Given flux at constant temperature
- 8 Given total flux
- 9 Given decay length
- 10 Leakage b.c.

GAMMAE is the secondary electron emission coefficient (when used), and GAMMAI the ratio of specific heats. If LBNDUSR=T, then the code uses *both* the b.c. from `b2mn.dat` and `b2.boundary.parameters`. Be careful then that b.c. are not applied twice!

Limiter b2.boundary.parameters (part I)

```

&boundary
NBC=10,
BCCHAR=      'S',      'S',      'S',      'N',      'N',      'W',      'E',      'W',      'E',      'N',
BCPOS=       -1,       -1,       -1,       24,       24,       -1,       96,       34,       30,       19,
BCSTART=      24,        0,       72,        0,       34,       -1,       -1,       20,       20,       31,
BCEND=        71,       23,       95,       30,       95,       24,       24,       23,       23,       33,
BCENE=         8,        9,        9,        9,        9,        3,        3,        3,        3,        9,
BCENI=         8,        9,        9,        9,        9,        3,        3,        3,        3,        9,
BCPOT=         0,        0,        0,        0,        0,        3,        3,        3,        3,        0,
ENEPAR=       2e6,      0.01,      0.01,      0.01,      0.01,      0.90,      0.90,      0.90,      0.90,      0.01,
ENEPAR(2,2)=1e2, 1e2, 1e2, 1e2, ENEPAR(10,2)=1e2,
ENIPAR=       2e6,      0.01,      0.01,      0.01,      0.01,      2.50,      2.50,      2.50,      2.50,      0.01,
POTPAR=        0.00,      0.00,      0.00,      0.00,      0.00,      0.00,      0.00,      0.00,      0.00,      0.00,
ENIPAR(2,2)=1e2, 1e2, 1e2, 1e2, ENIPAR(10,2)=1e2, POTPAR(1,2)=10*0,
BCCON(0,1)    =         8,         1,      CONPAR(0,1,1) = 0e19, 3e19,
BCCON(0,2)    =        10,        10,      CONPAR(0,2,1) =-1e-2, -1e-3,
BCCON(0,3)    =        10,        10,      CONPAR(0,3,1) =-1e-2, -1e-3,
BCCON(0,4)    =        10,        10,      CONPAR(0,4,1) =-1e-2, -1e-3,
BCCON(0,5)    =        10,        10,      CONPAR(0,5,1) =-1e-2, -1e-3,
BCCON(0,6)    =         0,         3,      CONPAR(0,6,1) = 0.00, 0.00,
BCCON(0,7)    =         0,         3,      CONPAR(0,7,1) = 0.00, 0.00,
BCCON(0,8)    =         0,         3,      CONPAR(0,8,1) = 0.00, 0.00,
BCCON(0,9)    =         0,         3,      CONPAR(0,9,1) = 0.00, 0.00,
BCCON(0,10)   =        10,        10,      CONPAR(0,10,1)=-1e-2, -1e-3,

```



Limiter b2.boundary.parameters (part II)

```
BCMOM(0,1)   =      4,      4,      MOMPAR(0,1,1) = 0.00,  0.00,  mompar(0,1,2)=2*1e5,
BCMOM(0,2)   =      4,      4,      MOMPAR(0,2,1) = 0.00,  0.00,  mompar(0,2,2)=2*1e5,
BCMOM(0,3)   =      4,      4,      MOMPAR(0,3,1) = 0.00,  0.00,  mompar(0,3,2)=2*1e5,
BCMOM(0,4)   =      4,      4,      MOMPAR(0,4,1) = 0.00,  0.00,  mompar(0,4,2)=2*1e5,
BCMOM(0,5)   =      4,      4,      MOMPAR(0,5,1) = 0.00,  0.00,  mompar(0,5,2)=2*1e5,
BCMOM(0,6)   =      3,      3,      MOMPAR(0,6,1) = 0.00,  1.00,  mompar(0,6,2)=2*0,
BCMOM(0,7)   =      3,      3,      MOMPAR(0,7,1) = 0.00,  1.00,  mompar(0,7,2)=2*0,
BCMOM(0,8)   =      3,      3,      MOMPAR(0,8,1) = 0.00,  1.00,  mompar(0,8,2)=2*0,
BCMOM(0,9)   =      3,      3,      MOMPAR(0,9,1) = 0.00,  1.00,  mompar(0,9,2)=2*0,
BCMOM(0,10)  =      4,      4,      MOMPAR(0,10,1)= 0.00,  0.00,  mompar(0,10,2)=2*1e5,
GAMMAI=3.0,GAMMAE=0.5,LBNDUSR=F,
NNISO  =  1, NXISO1 = 31, NXISO2 = 33, NYISO1 = 20, NYISO2 = 24,
NIISO  =  2*1.0E12, TEISO=1., TIISO=1., PHIISO=0./
```

Here we define NNISO(=1) dead region(s) which extend(s) in the computational rectangle (*NXISO1*(:), *NYISO1*(:)) to (*NXISO2*(:), *NYISO2*(:)).



Relevant switches and diagnostics

A few switches of use for special geometries:

b2mwti_jxi Poloidal index of the inner midplane

b2stbc_coreregn2 Index of the second core boundary condition in **b2ah.dat** and **b2mn.dat**. Similar to **b2stbc_coreregn0**.

b2agfs_periodic_bc Turns on diverted island geometry

b2agfs_nncut Used to overwrite the number of X-points **b2ag** finds

b2agfs_topcut2, ... Used to overwrite the **topcut** value from **b2ag** corresponding to the second X-point. Similar to **b2agfs_topcut**.

When running a double-null case, one should use **energy_analysis_dn** instead of **energy_analysis**. One should also do a **setenv NREG 8** so that all the regions can be followed with the usual run-time scripts.

A note about resizing non-standard meshes with **b2yt**: when setting the sizes **nx**, **nx1**, **ny** and **ny1** of a mesh, bear in mind that the guard cells cannot be rescaled. Thus taking a 90×24 double-null mesh and coarsening it by a factor of two in both directions gives a 46×12 mesh!